Java reflection, proxies, and annotations

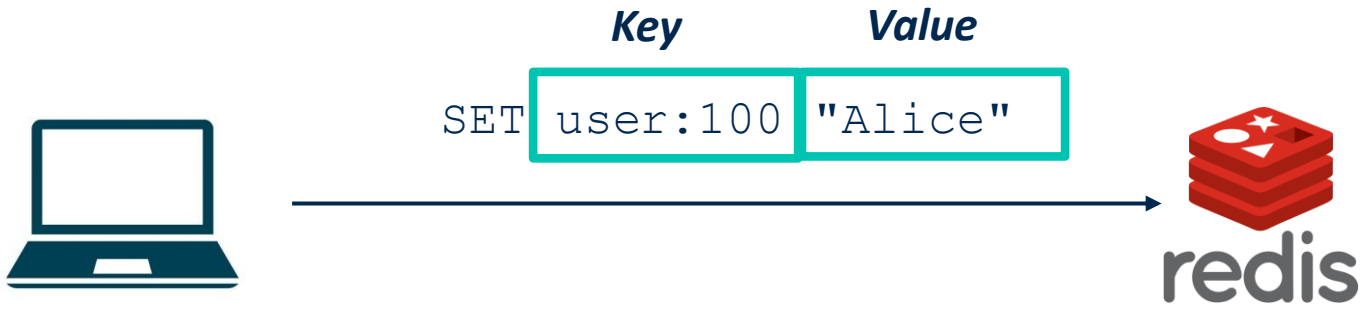
Tapti Palit

Next few topics

How do we design frameworks and libraries?

Use case: saving records in Redis

Redis is an in-memory key-value store



Use case: saving records in Redis

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Use case: saving records in Redis

Redis is an in-memory key-value store



Use case: Redis persistence from HW1

Can use CLI or library to communicate with Redis



SET mykey ”Hello, Redis!"



redis

```
Oct 13 10:58:39 ubuntu2404 systemd[1]: Starting redis-server.service - Advanced>
Oct 13 10:58:39 ubuntu2404 systemd[1]: Started redis-server.service - Advanced >

ubuntu@ubuntu2404:~$ redis-cli
127.0.0.1:6379> ping
PONG
127.0.0.1:6379> set mykey "Hello, Redis!"
OK
127.0.0.1:6379> get mykey
"Hello, Redis!"
127.0.0.1:6379> del mykey
(integer) 1
127.0.0.1:6379> exists mykey
(integer) 0
127.0.0.1:6379>
```

Use case: Redis persistence from HW1

Can use CLI or library to communicate with Redis



SET mykey "Hello, Redis!"



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Jedis

release v8.0.1 | maven-central v8.1.0-beta1 | javadoc 8.0.1 | license MIT | codecov 83% | chat 709 online

What is Jedis?

Jedis is a Java client for Redis designed for performance and ease of use.

Are you looking for a high-level library to handle object mapping? See redis-om-spring!

Potential design

- Design a `RedisDB` class that encapsulates all Redis persistence functionality
- `RedisDB` class internally uses a `Jedis` object from the jedis library
- Expose `persistAll` and `loadAll` methods to persist and load Student objects

```
class Student {
    // ... Fields and methodss
}


class RedisDB {
    Jedis jedis;

    RedisDB() { jedis = ...; }

    public void persistAll(Student student) {
        Map<String, String> studentMap;
        studentMap.put(“studentName”,
student.getName());
        studentMap.put(…, …);
        // ... All relevant fields
        jedis.hset(student.getId(), studentMap);
    }

}
```

Goal

- Design classes for social media platform
  - User, Post, Comment, Feed, Platform, Scheduler, Analytics, Message, Notification
- Add methods to the `RedisDB` class to persist all of them to a Redis database
- How to design/implement this?

Goal: reuse persistence code

- I wrote the persistence code once, why should I have to write it again?

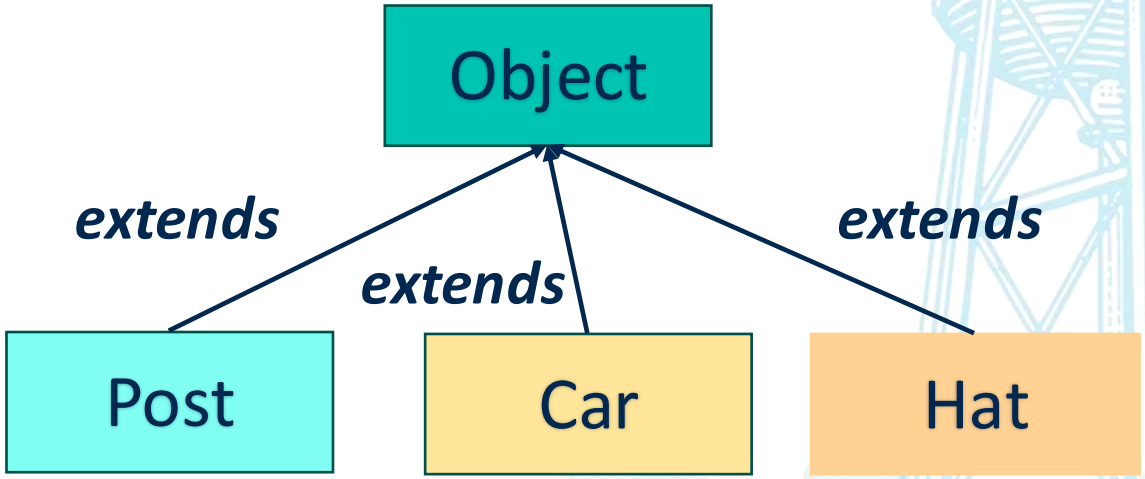
Goal: reuse persistence code

- Two pieces to this puzzle
  - Reflection
  - Annotation



Background

- In Java, all classes inherit the Object class implicitly



- Every instance method call's first implicit argument is the object it is being applied to
  - `Student s = new Student();`
    `s.getName(); // implicit first argument is s`

Reflection and metaprogramming

- Ability of a program to inspect and manipulate its own structure and behavior at runtime **within the same language environment**
- Can (dynamically) instantiate classes given a **class name** and invoke methods and constructors on it
- Purpose
  - Access and modify classes, methods, fields, and constructors dynamically
  - Facilitate frameworks, libraries, and tools (e.g., Spring, Hibernate)

Reflection and metaprogramming

- Types of Reflection
  - Compile time (C++ [https://en.cppreference.com/w/cpp/keyword/reflexpr.html])
  - Runtime (Java, JavaScript)

Java reflection

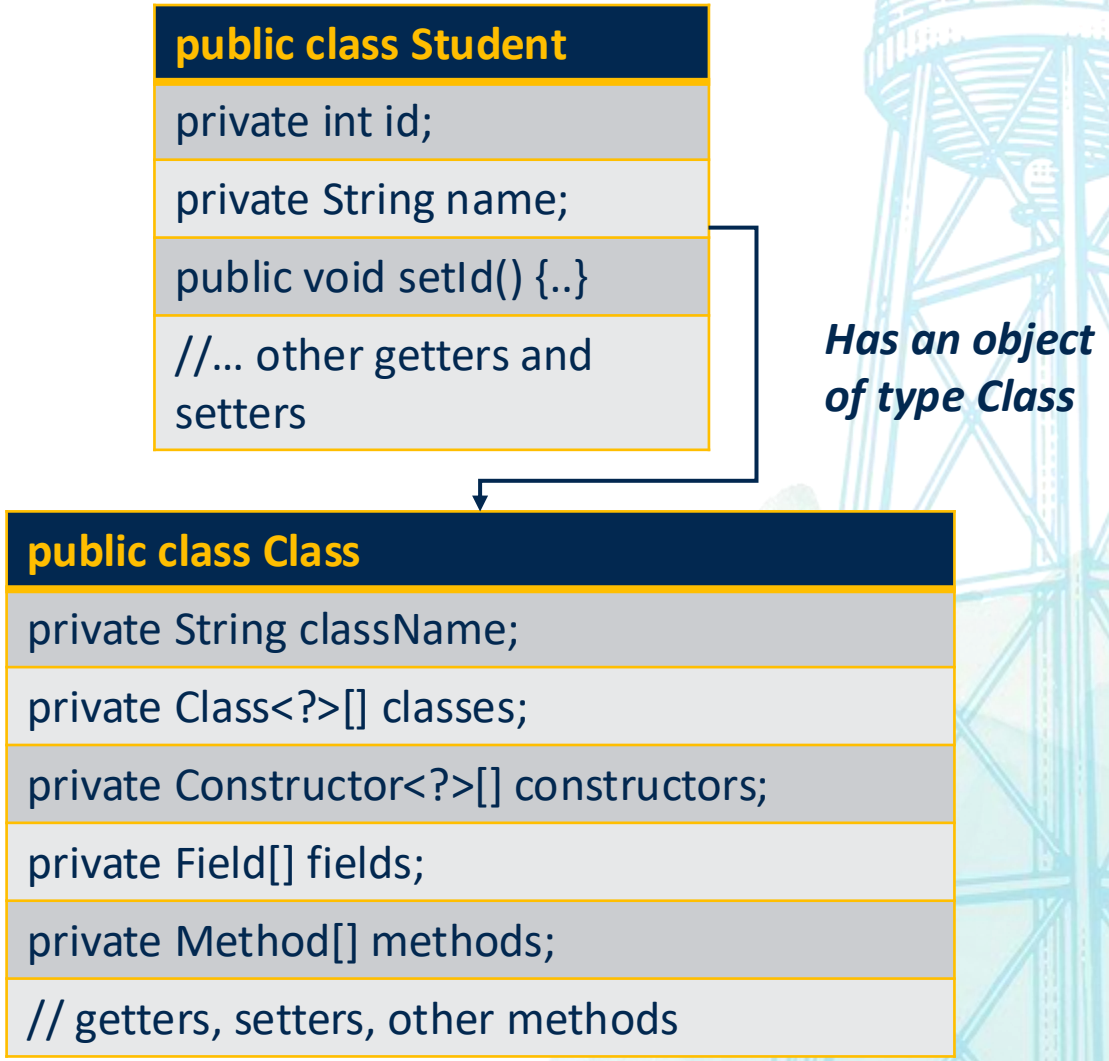
- Provides APIs to inspect and manipulate the classes, methods, fields, and so on…
- Note: reflection bypasses encapsulation
  - But (hopefully) for greater good!!

Key classes

- `java.lang.Class`: Represents a class or interface
- Every class definition in our program would have a Class object that Java compiler automatically creates

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Key classes

- java.lang.Class: Represents a class or interface
- Every class definition in our program would have a Class object that Java compiler automatically creates

public class Student
private int id;
private String name;
public void setId() {..}
//… other getters and setters

Class object
private String className; **// "Student"**
private Constructor<?>[] constructors; **// [default_cons]**
private Field[] fields; **[field_obj_id, field_obj_name]**
private Method[] methods; **[//all getters and setters]**
// getters, setters, other methods

Key classes

- `java.lang.reflect.Method`: Represents a class method
- `java.lang.reflect.Field`: Represents a class field
- `java.lang.reflect.Constructor`: Represents a constructor
- … what can we do with these objects of type Class, Method, Field, etc?

Reflection example

- E.g., using reflection to dynamically invoke all getter methods in an object

```
class Car {
    private String model;
    private int year;
    public String getModel() {return model;}
    public int getYear() { return year; }
    public Car(String model, int year) {...}
}

class MyApp {
    public static void main(String[] args) {
        Car c = new Car("Toyota", 2019);
        Class clazz = c.getClass();
        for (Method m: clazz.getDeclaredMethods()) {
            if (m.getName().startsWith("get")) {
                Object result = m.invoke(c);
                S.o.p(result);
            }
        }
    }
}
// Prints "Toyota" and then "2019"
```

Same reflection code works on all objects

- Using reflection on another object type
- … what did we achieve?

```
class Post {
     private String authorName;
     private String content;
     private int replyCount;
     public int getAuthorName() { return authorName; }
     public int getContent() { return content; }
     public Integer getReplyCount() { return replyCount; }
     public Post(String authorName, String content,
          int replyCount) {
          this.authorName = authorName;
          this.content = content;
          this.replyCount = replyCount;
     }
}

class MyApp {
     public static void main(String[] args) {
          Post p = new Post("Tapti", "Welcome to ECS 160",
0);
          Class clazz = p.getClass();
          for (Method m: clazz.getDeclaredMethods()) {
              if (m.getName().startswith("get") {
                Object result = m.invoke(p);
                S.o.p(result);
                }
          }
     }
}
// Prints "Tapti", "Welcome to ECS 160", and then "0"
```

Reflection API

- `Class.forName(<classname>)` – loads the Class object provided a fully qualified class name (`com.ecs160.MyClass`)
- `<obj>.getClass()` - Gets the `Class` object for the object `obj`
- `<class_obj>.getMethods()` – Get all method objects of type `Method` for the Class object `class_object`
- `<class_obj>.getMethod(<method_name>)` - Get the `Method` object for the method having `method_name`

Why reflection?

- Goal: reuse `persistAll` logic, independent of the object type

```
class Post { // fields of post}

class RedisDB {
    Jedis jedisSession;
    static int id;
    static {
        jedisSession = new Jedis(“localhost”,
6379);
        id = 0;
    }

    public void persistAll(Post post) {
        Map<String, String> postMap;
        postMap.put(“authorName”,
post.getAuthor());
        postMap.put(…, …);
        // ... All relevant fields
        jedis.hset(id++, postMap);
    }

}
```

Why reflection?

```
class Post { }

class RedisDB {
    Jedis jedisSession;
    static int id;
    static {
        jedisSession = new Jedis(“localhost”,
6379);
        id = 0;
    }

    public void persistAll(Post post) {
        Map<String, String> postMap;
        postMap.put(“authorName”,
post.getAuthor());
        postMap.put(…, …);
        // ... All relevant fields
        jedis.hset(id++, postMap);
    }

}
```

```
class Post { }
class Car {}

class RedisDB {
     Jedis jedisSession;
     static int id;
     static {
          jedisSession = new Jedis(“localhost”, 6379);
          id = 0;
     }

     public void persistAll(Object obj) {
          Map<String, String> postMap;
          Class c = obj.getClass();
          for (Field f: c.getDeclaredFields()) {
             String fieldname = f.getName();
             fieldVal.setAccessible(true);
             Object fieldVal = f.get(obj);
             postMap.put(fieldname, fieldVal);
          }
          jedis.hset(id++, postMap);
     }
}

Post p = new Post(“Hello world”, “1/23/2025”);
Car c = new Car(“BMV”);
RedisDB db = new RedisDB();
db.persistAll(p);
db.persistAll(c);
```

Reflection recap

- `java.lang.reflect.Class`
  - An object of this type encapsulates the class information for a particular class
  - `Class clazz = Class.forName("com.ecs160.MyApp");`
  - `Class clazz = obj.getClass();`
- `java.lang.reflect.Method`
  - Invoke using `m.invoke(obj);`
- `java.lang.reflect.Field`
  - Invoke using `f.getVal(obj);`
- Reflection can bypass access modifiers using the `setAccessible(true)` method
- Invoke constructor to create a new Object
  - `Object o = c.getConstructor().newInstance();`

Loading all classes

- ```
  ClassLoader cl = ClassLoader.getSystemClassLoader();
  java.lang.reflect.Field classesField =
  ClassLoader.class.getDeclaredField("classes");
  classesField.setAccessible(true);
  Vector<Class<?>> classes = (Vector<Class<?>>)
  classesField.get(cl);
  for (Class<?> c : classes) {
    System.out.println(c.getName());
  }
  ```
  ```

Discussion: design problem

- Design a framework that invokes all methods in all classes whose names start with Test

Discussion: design problem

- Design a framework that invokes all methods in all classes whose names start with Test
- Pseudo-code
  - ```
    List<class> classes = classLoader.getClasses();
    for (Class clazz: classes) {
       if (clazz.getName().startsWith(“Test”) {
          Object o = clazz.constructor.invoke();
          for (Method m: clazz.getDeclaredMethods()) {
             m.invoke(o);
          }
       }
    }
    ```
    ```

Drawback of previous approach

- All fields saved
- What if we want only a subset of all fields persisted?
- Solution: annotations

```
class Post { }

class RedisDB {
     Jedis jedisSession;
     static int id;
     static {
          jedisSession = new Jedis(“localhost”, 6379);
          id = 0;
     }

     public void persistAll(Object obj) {
          Map<String, String> postMap;
          for (Field f: obj.getDeclaredFields()) {
             String fieldname = f.getName();
             Object fieldVal = f.get(obj);
             fieldVal.setAccessible(true);
             postMap.put(fieldname, fieldVal);
          }
          jedis.hset(id++, postMap);
     }

}

Post p = new Post(“Hello world”, “1/23/2025”);
Car c = new Car(“BMV”);
RedisDB db = new RedisDB();
db.persistAll(p);
db.persistAll(c);
```

Java annotations

- Annotations are “metadata” added to Java code
- They have no direct impact on code execution at runtime
- Java provides some annotations, programmer can define more
- Syntax: `@Annotation_name`
- Widely used in popular frameworks

Annotation targets

- Annotations can be applied to
  - Classes
  - Fields
  - Methods
  - … many other program elements (Check https://docs.oracle.com/javase/tutorial/java/annotations/basics.html)

Annotations for compiler checks

- Additional information for the compiler
  - Detect errors
  - Suppress warnings
- Common predefined annotations
  - @Override, @Deprecated, @SuppressWarnings

```
class Vehicle {
    public void start() {
        System.out.println(“Vehicle starts”);
    }
}

class Bicycle extends Vehicle {
    @Override
    public void start() { ... }

    @Override // COMPILER ERROR!
    public void stop() { ... }
}
```

Few common predefined annotations

- `@Override`
  - The annotated method must override a parent method. If not, results in compiler error
- `@Deprecated`
  - The annotated method is deprecated and will show a compiler warning if used
- `@SuppressWarnings`
  - Compiler errors for the annotated entity are suppressed

Defining new annotations

- Annotation definition must begin with @interface
- Annotations can have multiple fields
  - *Annotation type element* declarations
- Each field has a constructor
  - Constructors can have default values
- Fields can consist of arrays

```
@interface MyAnnotation {
   String author();
   String date();
   int currentRevision() default 1;
   String lastModified() default "N/A";
   String lastModifiedBy() default "N/A";
}
```

```
@MyAnnotation (
    author = "John Doe",
    date = "3/17/2002",
 )
 public class Car extends Vehicle {
    // ...
 }
```

Recall: compilation toolchain



MyApp.java

javac

Some annotations retained only in the compiler

```
Compiled from "MyApp.java"
public class MyApp {
  public MyApp();
    Code:
      0: aload_0
      1: invokespecial #1                  // Method
java/lang/Object."<init>":()V
      4: return

  public static void main(java.lang.String[]);
    Code:
      0: getstatic     #7                  // Field
java/lang/System.out:Ljava/io/PrintStream;
      3: ldc           #13                 // String Hello!
      5: invokevirtual #15                 // Method
java/io/PrintStream.println:(Ljava/lang/String;)V
      8: return
}
```

MyApp.class

INTERPRETER

JVM (Java Virtual Machine)

Just-in-Time Compiler

[java]

Some annotations retained at runtime

scratch.asm

```
format PE console

push ebp
mov  ebp, esp

mov  eax, 10

mov  esp, ebp
pop  ebp
ret
```

Machine code

Reflection can access annotations

- Can check if annotation is present
  - `Class clazz = … ;`
    `clazz.isAnnotationPresent(MyAnnotation.class)`
  - `Field field = …;`
    `field.isAnnotationPresent(…);`
  - Same for `Method`
- Can get the annotation
  - `MyAnnotation myAnnotation = clazz.getAnnotation(MyAnnotation.class)`
- Can get the values of the annotation element fields
- **In class demo**

Drawback of previous approach

- All fields saved
- What if we want only a subset of all fields persisted?
- Solution: annotations

```
class Post { }

class RedisDB {
     Jedis jedisSession;
     static int id;
     static {
          jedisSession = new Jedis(“localhost”, 6379);
          id = 0;
     }

     public void persistAll(Object obj) {
          Map<String, String> postMap;
          for (Field f: obj.getDeclaredFields()) {
             String fieldname = f.getName();
             Object fieldVal = f.get(obj);
             postMap.put(fieldname, fieldVal);
          }
          jedis.hset(id++, postMap);
     }

}

Post p = new Post(“Hello world”, “1/23/2025”);
Car c = new Car(“BMV”);
RedisDB db = new RedisDB();
db.persistAll(p);
db.persistAll(c);
```

Full solution

- Create an annotation @Persistable with Runtime retention policy
- Annotate only some fields
- When persisting the fields, check if the annotation is present
  - Only persist if the annotation is present

```
@Retention(RetentionPolicy.RUNTIME)
public @interface Persistable {
}

class Post {
     @Persistable
     private String content;
     private Integer tempVal;
}

class RedisDB {
     // … set up Jedis Session
     public void persistAll(Object obj) {
          Map<String, String> postMap;
          for (Field f: obj.getDeclaredFields()) {
             if (f.isAnnotationPresent(Persistable.class) {
                f.setAccessible(true);
                String fieldname = f.getName();
                Object fieldVal = f.get(obj);
                postMap.put(fieldname, fieldVal);
             }
          }
          jedis.hset(id++, postMap);
     }

}
```

Reflection (recap)

- Create an annotation @Persistable with Runtime retention policy
- Annotate only some fields
- When persisting the fields, check if the annotation is present
  - Only persist if the annotation is present

```
@Retention(RetentionPolicy.RUNTIME)
public @interface Persistable {
}

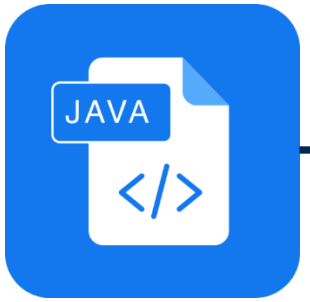
class Post {
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     private String content;
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          for (Field f: obj.getDeclaredFields()) {
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                String fieldname = f.getName();
                Object fieldVal = f.get(obj);
                postMap.put(fieldname, fieldVal);
             }
          }
          jedis.hset(id++, postMap);
     }

}
```

Recall: compilation toolchain

Compile Time



MyApp.java

javac

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Compiled from "MyApp.java"
public class MyApp {
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    Code:
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  public static void main(java.lang.String[]);
    Code:
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}
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MyApp.class

Some annotations retained only in the compiler

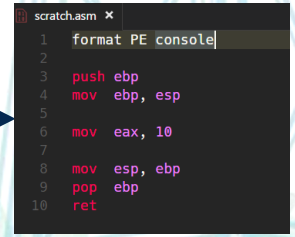
Run Time

INTERPRETER



Just-in-Time Compiler

[java]



Machine code

Some annotations retained at runtime

Annotations and retention policies

- The `@Retention` annotation indicates the retention policy for the annotation
- `@Retention(RetentionPolicy.SOURCE)` – only present in the source file
- `@Retention(RetentionPolicy.CLASS)` – only present in the class file
- `@Retention(RetentionPolicy.RUNTIME)` – present at runtime
- Reflection can only access annotations with `RetentionPolicy.RUNTIME`

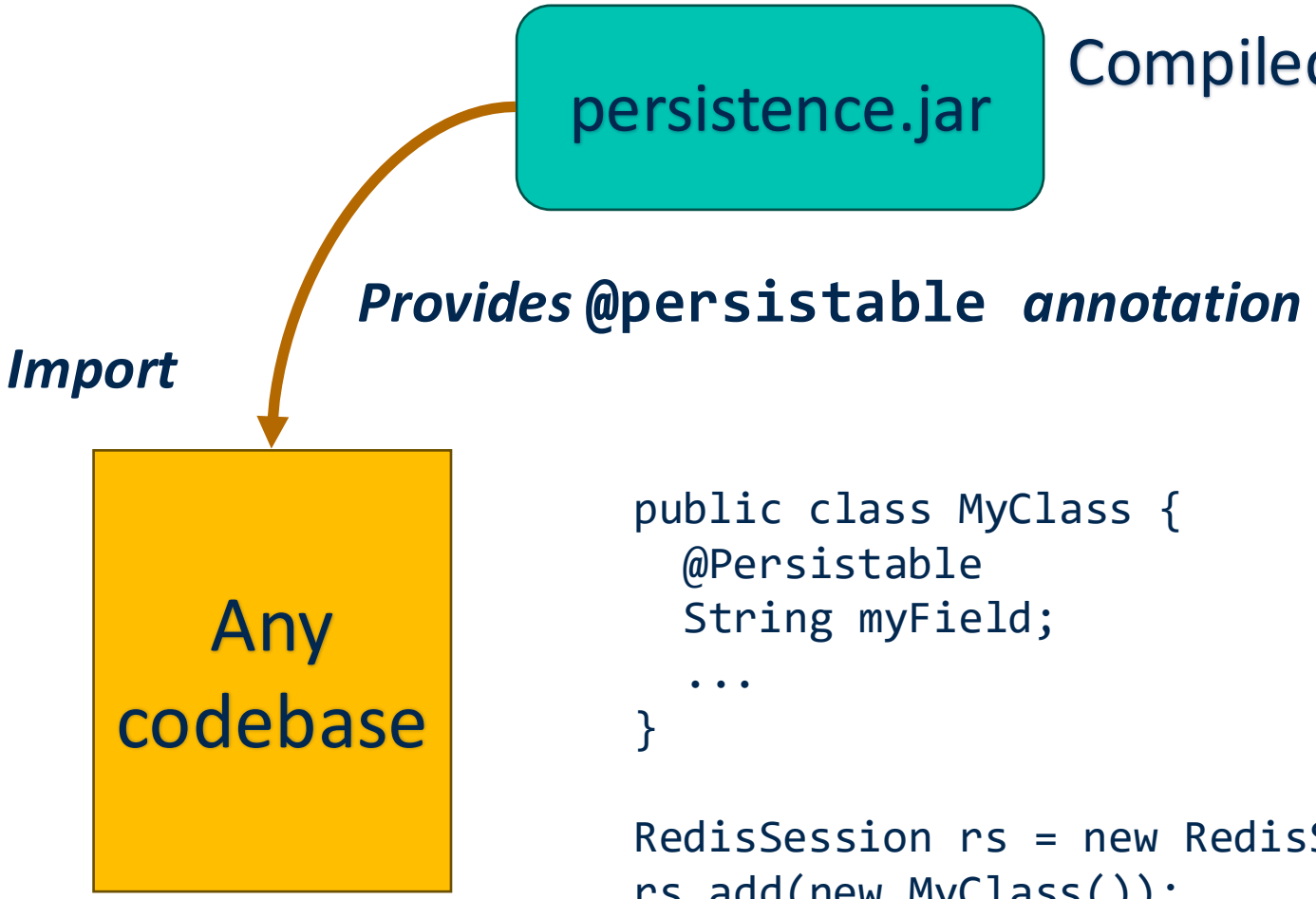
```
@Retention(RetentionPolicy.RUNTIME)
@interface MyAnnotation {
   String author();
   String date();
   int currentRevision() default 1;
   String lastModified() default "N/A";
   String lastModifiedBy() default "N/A";
}
```

Why would we want to create our own annotations visible only at the source code / class level?

Reflection + annotations: summary

- What did we achieve?
  - The ability to persist ***any*** object, provided it is annotated
  - The persistence framework ***does not need*** to know the classes that can be persisted
  - The persistence logic is ***fully decoupled*** from the application's data model

Distribute persistence framework as a library




```
public class MyClass {
  @Persistable
  String myField;
  ...
}

RedisSession rs = new RedisSession();
rs.add(new MyClass());
rs.persistAll();
```

Object relational mapping (ORM) frameworks

- Hibernate: allows the programmer to annotate a class with `@Entity`
- Provides an API `save()` that accepts an object of a class annotated with `@Entity` and saves it in the database
- The framework *abstracts* the SQL logic
- **In-class demo**




```
@Entity
public class Car {
        @Id
        @GeneratedValue(strategy= GenerationType.IDENTITY)
        private Long id;

        private String name;
        private String email;
}
```

Unit testing frameworks

- JUnit: allows the programmer to annotate a class with `@Test`
- Can specify pre- and post-operations
- All methods of the class automatically executed using Reflection
- The framework *abstracts* the invocation logic of the tests




```
public class CalculatorTest {
        @Test
        public void testAdd() {
                calculator = new Calculator();
                int result = calculator.add(2, 3);
                assertEquals(5, result, "2 + 3 should equal 5");
        }
}
```

… and more

- Microservice frameworks
  - Spring Boot
- MVC frameworks
  - Spring MVC
- Logging frameworks
  - Log4j, SL4J
- … and create other design patterns such as ***Inversion of Control (IoC)***









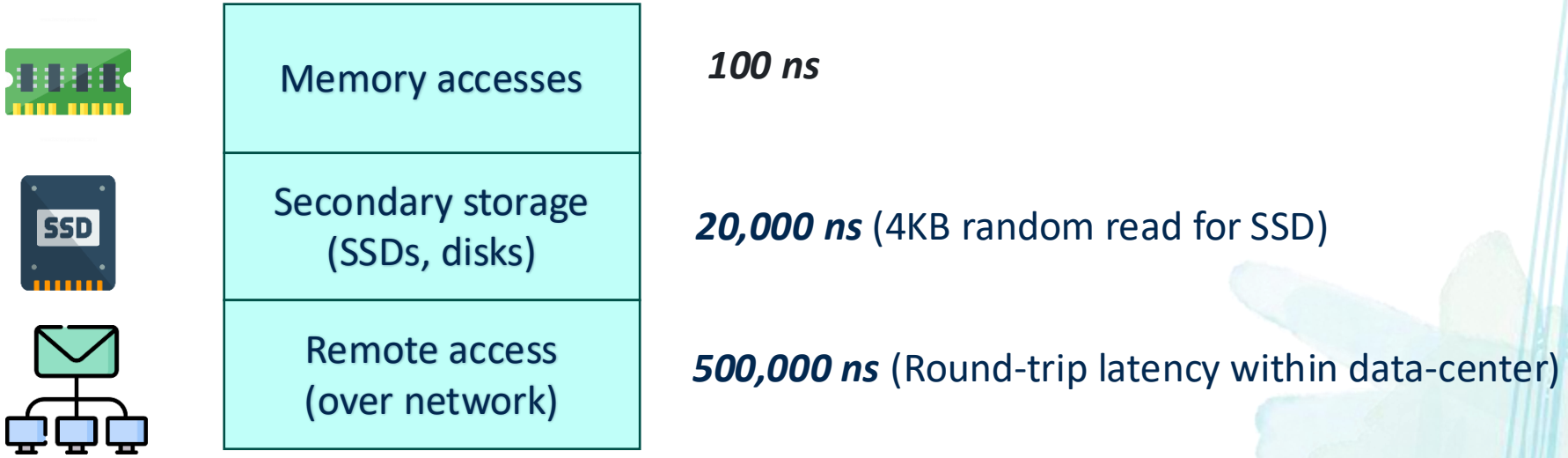
Downsides of reflection

- Cannot *generate* new methods, fields, etc.
- Reflection causes full loss of type safety
  - If you do `s.setName(new Integer(123));` compiler will catch it at compile time
  - If you do `setNameMethod.invoke(s, new Integer(123));` it will throw an exception at runtime
- Performance is slower than directly accessing the field/method
  - In some cases, it is okay
    - Database persistence, network communications
    - Why?

Latency hierarchy

- CPU register << DRAM memory << SSDs << hard disks << network

https://static.googleusercontent.com/media/sre.google/en//static/pdf/rule-of-thumb-latency-numbers-letter.pdf



- Adding a few extra memory accesses for reflection is not noticeable for disk and network operations

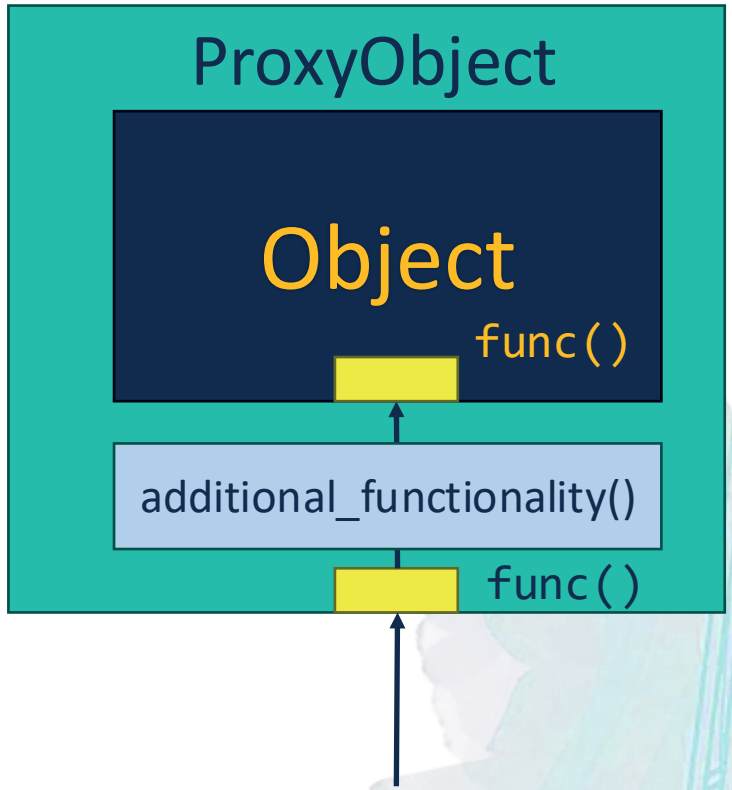
Dynamic Proxies

Tapti Palit

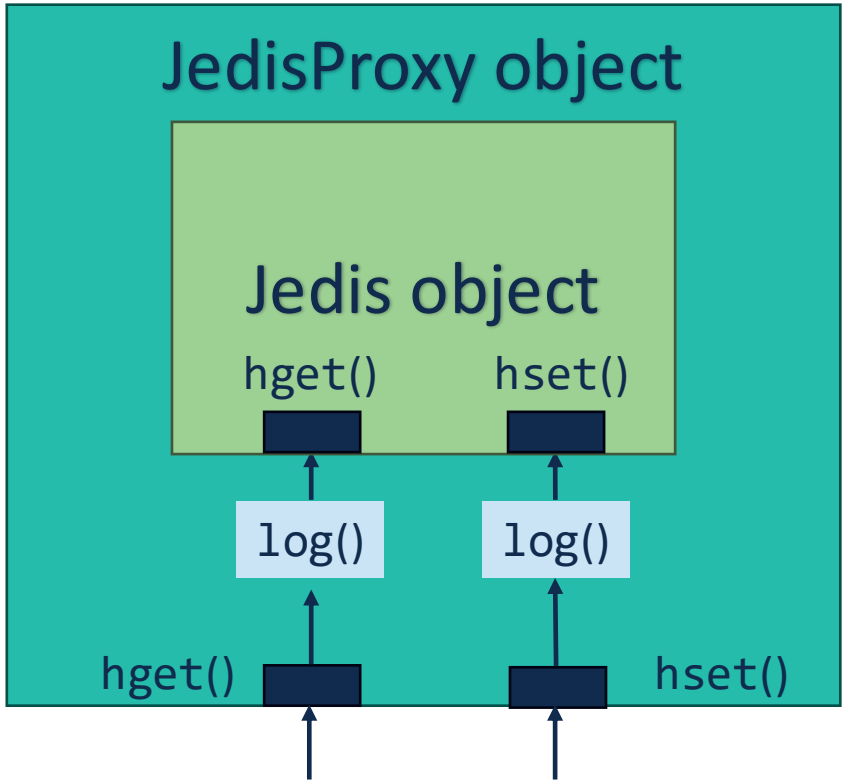


Proxy design pattern - what is a proxy object?

- A proxy is a wrapper around the original/target object
- The user accesses the proxy object instead of the original target object
- The proxy object typically
  - Performs some additional logic
  - Then forwards the request to the target object
    - ***Method interception***



Proxy object wraps and extends target class




```
class Jedis {
    public String hget(String id) { … }
    public void hset(String id, String val) { … }
}

class JedisProxy extends Jedis{
    private Jedis jedis; // Wrap jedis obj

    public log() { S.o.p(…); }

    public String hget(String id) {
        log();
        return jedis.hget(id);
    }
    public void hset(String id, String val) {
        log();
        jedis.hset(id, val);
    }
}
```

Proxies intercept the method invocations

What are the limitations?

- Explicitly create a class that wraps the original object type
- Explicitly create objects of this proxy class
- ***Limitation:*** proxy classes must be statically designed for each "proxyable" class
  - Duplicated logging logic

```
class SQL {
    public void executeQuery(String query) { … }
}
```

```
class SQLProxy extends Query{
    public log() { S.o.p(…); }

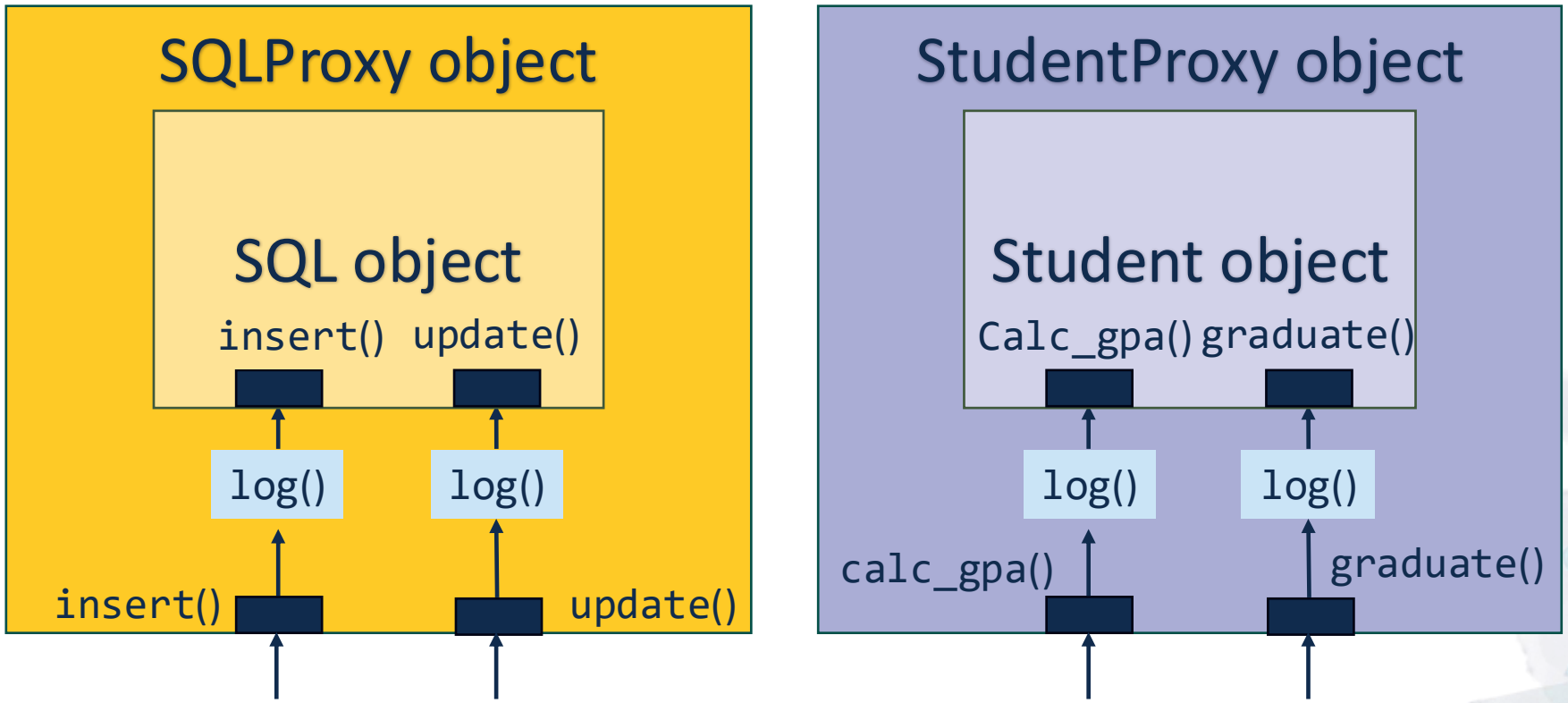
    public void executeQuery(String query) {
        log();
        query.executeQuery(query);
    }

}
```

```
SQL sql = …; // sql object
SQLProxy sqlProxy = new SQLProxy(sql);
```

Duplicated work

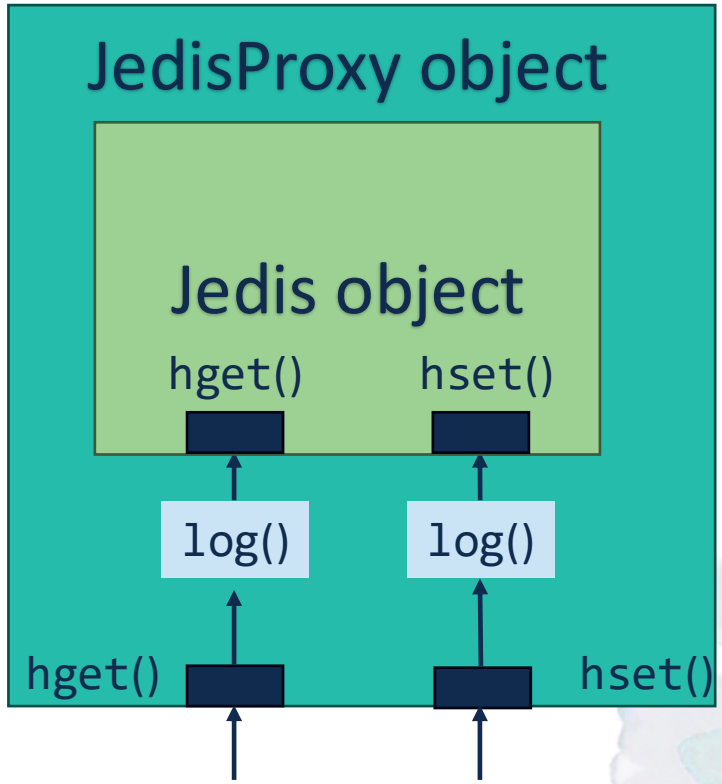
- Add logging to other classes

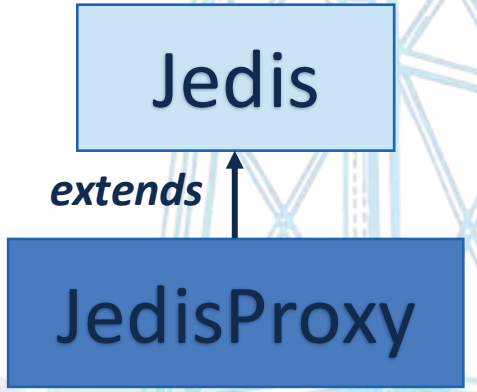


Duplicated work designing proxy classes for each class

Proxy goals

- Want to reuse the proxy functionality (logging, for example)
- Would be nice to ***dynamically*** create a ***subclass*** for the target object class that wraps any target object with the proxy functionality
- E.g. magical method which accepted the `log()` method and the `Jedis` class, and generated `JedisProxy` ***on the fly***





Proxy goals

- Want to reuse the proxy functionality (logging, for example)
- Would be nice to ***dynamically*** create a ***subclass*** for the target object class that wraps any target object with the proxy functionality
- E.g. magical method which accepted the `log()` method and generated `JedisProxy` ***on the fly***

```
Object createProxy(Object target, [FUNCTION
encapsulating the additional functionality]) {

    // 1. Create proxyClass which is a subclass
of target.getClass()
    // 2. This proxyclass will intercept all
method invocations on itself
    // 3. And invoke the additional
functionality and then retarget the method
invocation to the target object
    // 4. Create and return an object of
proxyClass
}
```

Runtime bytecode generation and manipulation

- ByteBuddy, Javassist libraries allow proxy creation via dynamic subclassing
  - ByteBuddy and Javassist hide bytecode manipulation complexities
  - Internally uses ASM library which gives complete bytecode generation/manipulation capabilities
- Java also has a dynamic proxy functionality
  - IMHO, it's unnecessarily complicated
  - Is an example of a language providing support for a design pattern!

Proxying using Javassist

- Create proxyClass which is a subclass of `target.getClass()`
- This `proxyclass` will intercept all method invocations on itself
- And invoke the additional functionality and then retarget the method invocation to the target object
- Create and return an object of `proxyClass`

```
Object createProxy(Object object) {


}
```



Proxying using Javassist

- **Create proxyClass which is a subclass of `target.getClass()`**
- This `proxyclass` will intercept all method invocations on itself
- And invoke the additional functionality and then retarget the method invocation to the target object
- Create and return an object of `proxyClass`

```
Object createProxy(Object object) {

 Class<?> clazz = object.getClass();

 ProxyFactory proxyFactory = new ProxyFactory();
 proxyFactory.setSuperclass(clazz);
 Class<?> proxyClass = proxyFactory.createClass();

 // We will see how to specify the additional
 // functionality in next slide
}
```

Proxying using Javassist

- Create proxyClass which is a subclass of `target.getClass()`
- This `proxyclass` will intercept all method invocations on itself
- And invoke the additional functionality and then retarget the method invocation to the target object
- **Create and return an object of `proxyClass`**

```
Object createProxy(Object object) {

 Class<?> clazz = object.getClass();

 ProxyFactory proxyFactory = new ProxyFactory();
 proxyFactory.setSuperclass(clazz);
 Class<?> proxyClass = proxyFactory.createClass();

 MethodHandler methodHandler = new LogHandler();
    // Additional functionality (more next slide)

 Object proxyObject =
proxyClass.getDeclaredConstructor().newInstance();

 ((javassist.util.proxy.Proxy)
proxyObject).setHandler(methodHandler);

 return proxyObject;
}
```

Proxying using Javassist

- Create proxyClass which is a subclass of `target.getClass()`
- **This `proxyclass` will intercept all method invocations on itself**
- **And invoke the additional functionality and then retarget the method invocation to the target object**
- Create and return an object of `proxyClass`

```
Object createProxy(Object object) {
 // -- snip
 MethodHandler methodHandler = new LogHandler();
    // Additional functionality (more next slide)
 // -- snip
}

class LogHandler extends MethodHandler {
    @Override
    public Object invoke(Object self,
        Method thisMethod,
        Method proceed,
        Object[] args) throws Throwable {
        log("accessing method" +
thisMethod.getName());
        return proceed.invoke(self, args);
    }
}
```

Javassist ensures that all method invocations of proxy object intercepted by* `invoke` *method

Proxying using Javassist

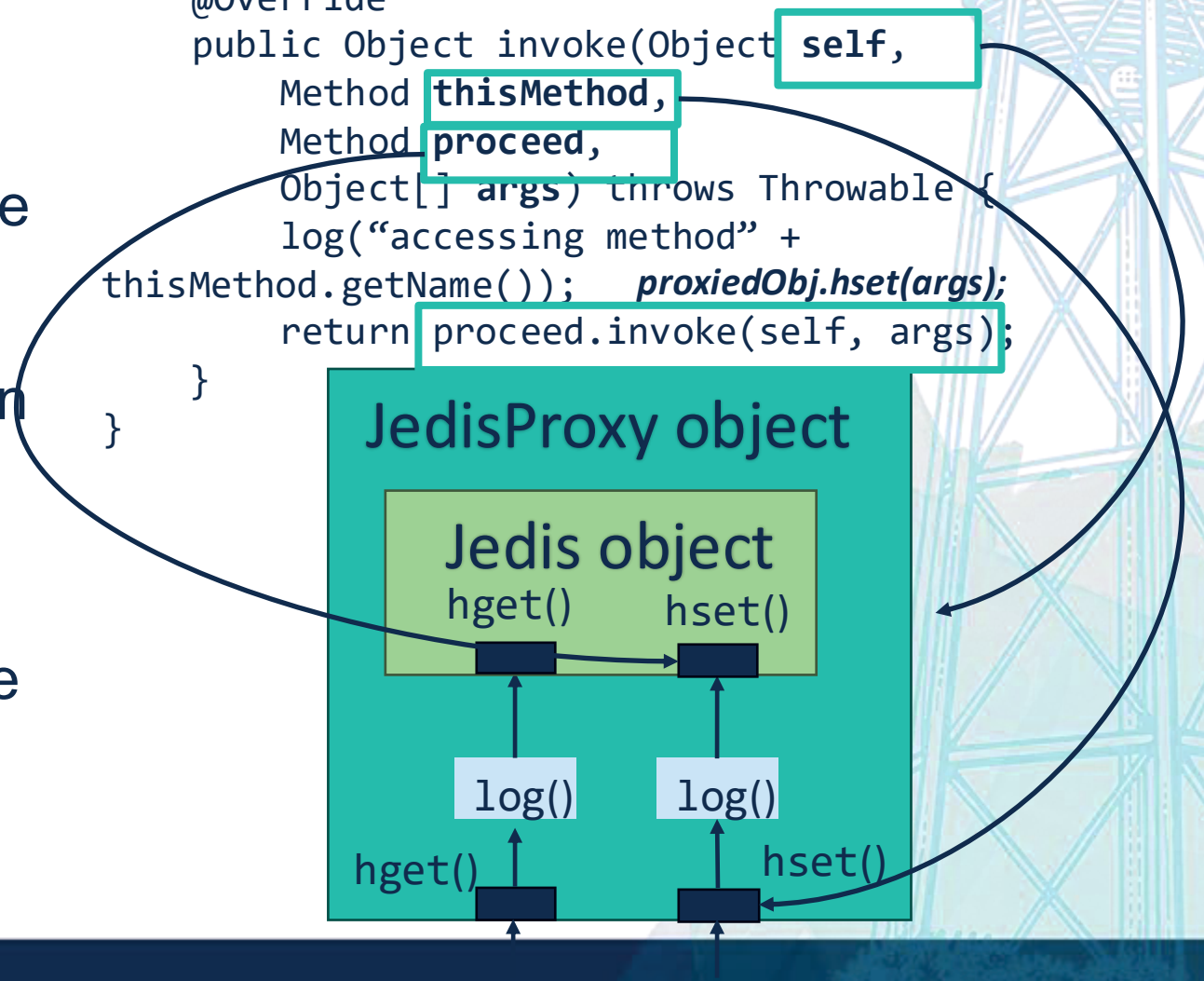
- Arguments to invoke method
  - `self` – the proxy object
  - `proceed` – the invoked method in the target object
  - `thisMethod` – the invoked method in the proxy object (generally not useful)
  - `args` – any arguments passed to the method invocation

```
class LogHandler extends MethodHandler {
    @Override
    public Object invoke(Object self,
        Method thisMethod,
        Method proceed,
        Object[] args) throws Throwable {
        log("accessing method" +
thisMethod.getName());
        return proceed.invoke(self, args);
    }
}
```

Proxying using Javassist

- Arguments to invoke method
  - `self` – the proxy object
  - `proceed` – the invoked method in the target object
  - `thisMethod` – the invoked method in the proxy object (generally not useful)
  - `args` – any arguments passed to the method invocation

```
class LogHandler extends MethodHandler {
    @Override
    public Object invoke(Object self,
        Method thisMethod,
        Method proceed,
        Object[] args) throws Throwable {
        log("accessing method" +
thisMethod.getName());   proxiedObj.hset(args);
        return proceed.invoke(self, args);
    }
}
```



Using dynamic proxies

```
Object createProxy(Object object) {
 Class<?> clazz = object.getClass();

 ProxyFactory proxyFactory = new ProxyFactory();
 proxyFactory.setSuperclass(clazz);

 MethodHandler methodHandler = new LogHandler();
 Class<?> proxyClass =
proxyFactory.createClass();
 Object proxyObject =
proxyClass.getDeclaredConstructor().newInstance()
;
 ((javassist.util.proxy.Proxy)
proxyObject).setHandler(methodHandler);
 return proxyObject;
}

class LogHandler {
  // -- snip
}
```

```
Jedis jedis = …;

Jedis jedisProxy = (Jedis) createProxy(jedis);
jedisProxy.hset(…);
```

```
Student student = …;

Student studentProxy =
              (Student)createProxy(student);
studentProxy.getName();
```

Demo

- https://github.com/davsec-teaching/javassist_demo



Real world use cases of proxies

- Sample use cases
  - Lazy loading
  - Mock testing



HIBERNATE

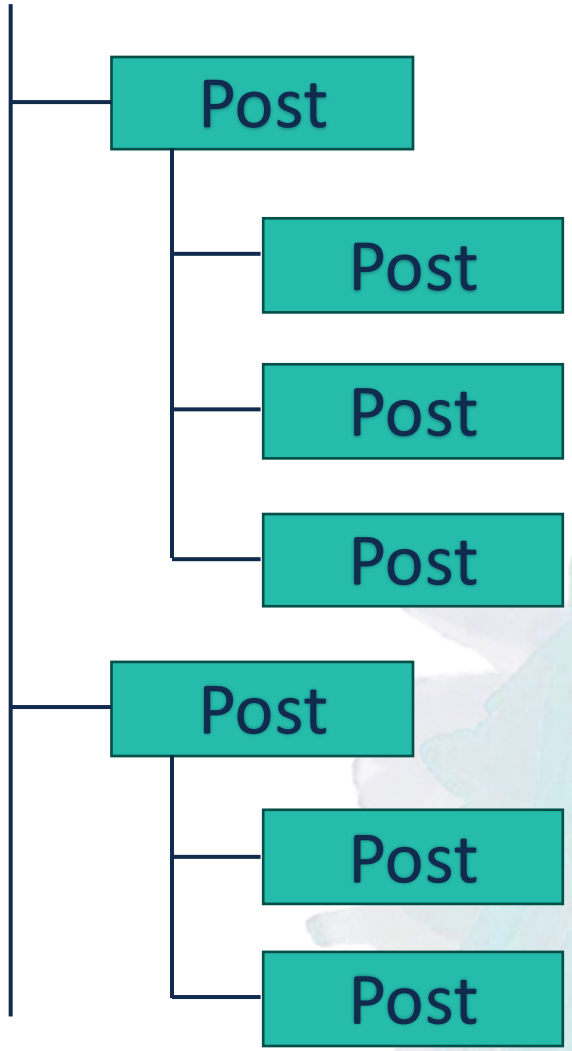


mockito

EASYMOCK

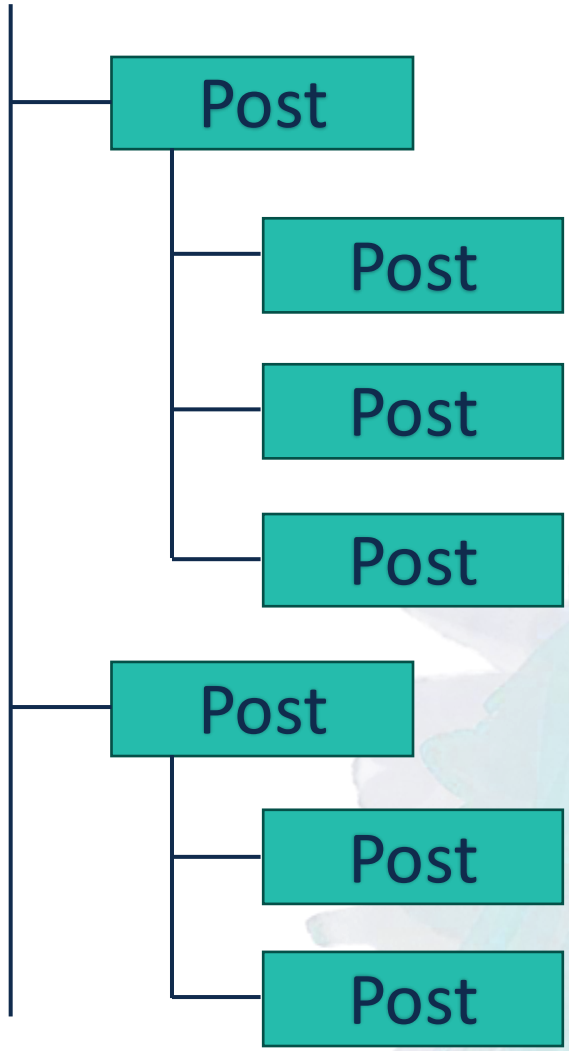
Loading posts

- Load a single post from the Redis database



Loading posts

- Load a single post from the Redis database
- High level overview
  - Load a post
  - For each reply
    - Load the reply



Loading posts from Redis

```
map = jedis.hgetAll(“3208”);
Post post = new Post();
post.setId(“3208”);
post.setCreatedAt(map.get(“createdAt”));
```

```
List<String> replies =
map.get(“childPosts).split(“,”);

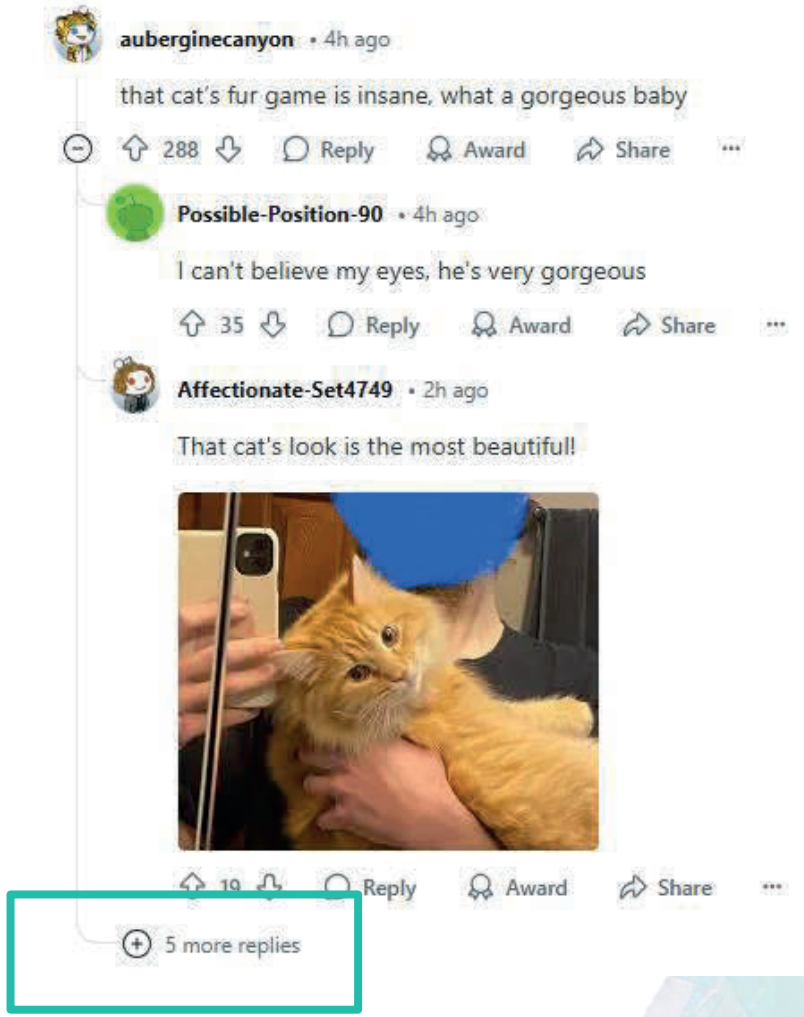
for (String replyId: replies) {
    replyMap = jedis.hgetAll(replyId);
    Post reply = new Post();
    reply.setId(replyId);
    reply.setCreatedAt(replyMap.get(“createdAt”);
    post.getReplies().add(reply);
}
```

```
127.0.0.1:6379> hgetall 3208
 1) "children"
 2) ""
 3) "QuoteCount"
 4) "0"
 5) "Author"
 6) "Author{handle='aparker.io', name='austin \xf0\x9f\x8e\x84'}"
 7) "Id"
 8) "3176"
 9) "ReplyCount"
10) "2"
11) "LikeCount"
12) "15"
13) "PostContent"
14) "bluesky brought to you by verisign"
15) "RepostCount"
16) "1"
17) "childPosts"
18) "3177,3188,"
127.0.0.1:6379>
```

All replies loaded when the post is loaded

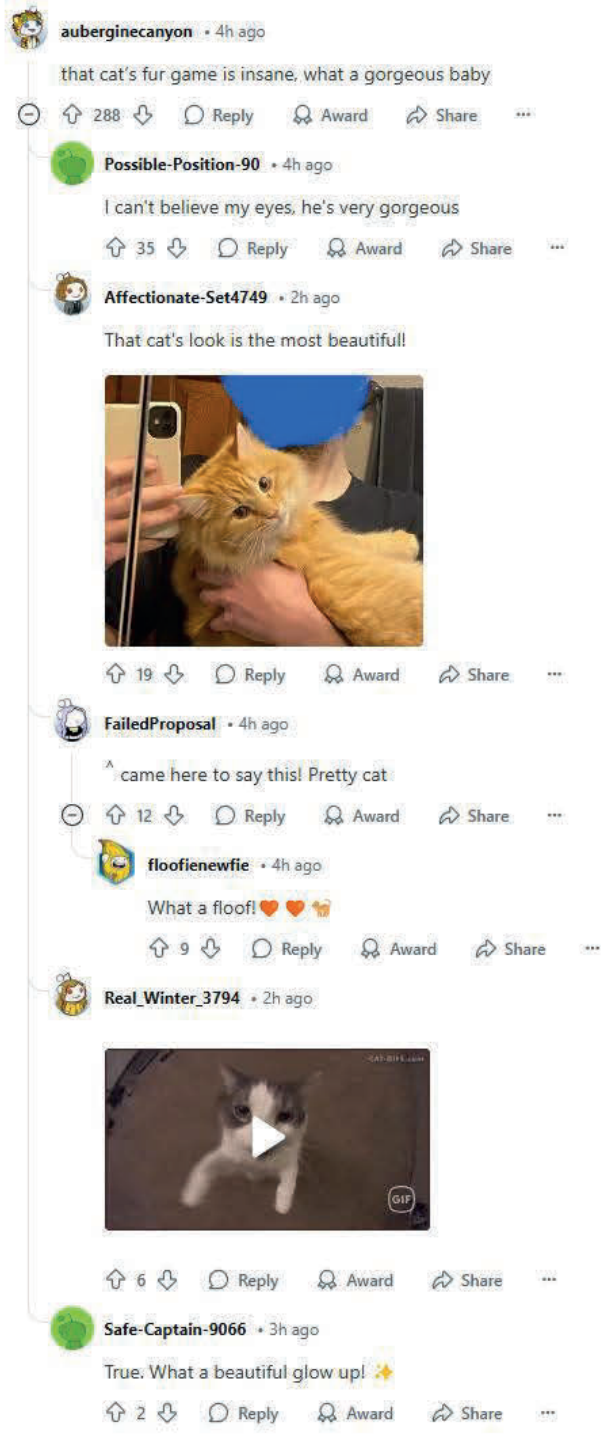
Lazy loading design pattern

- Common performance improvement
- An object's children (replies) are lazily loaded/fetched on-demand
- Improves UI responsiveness



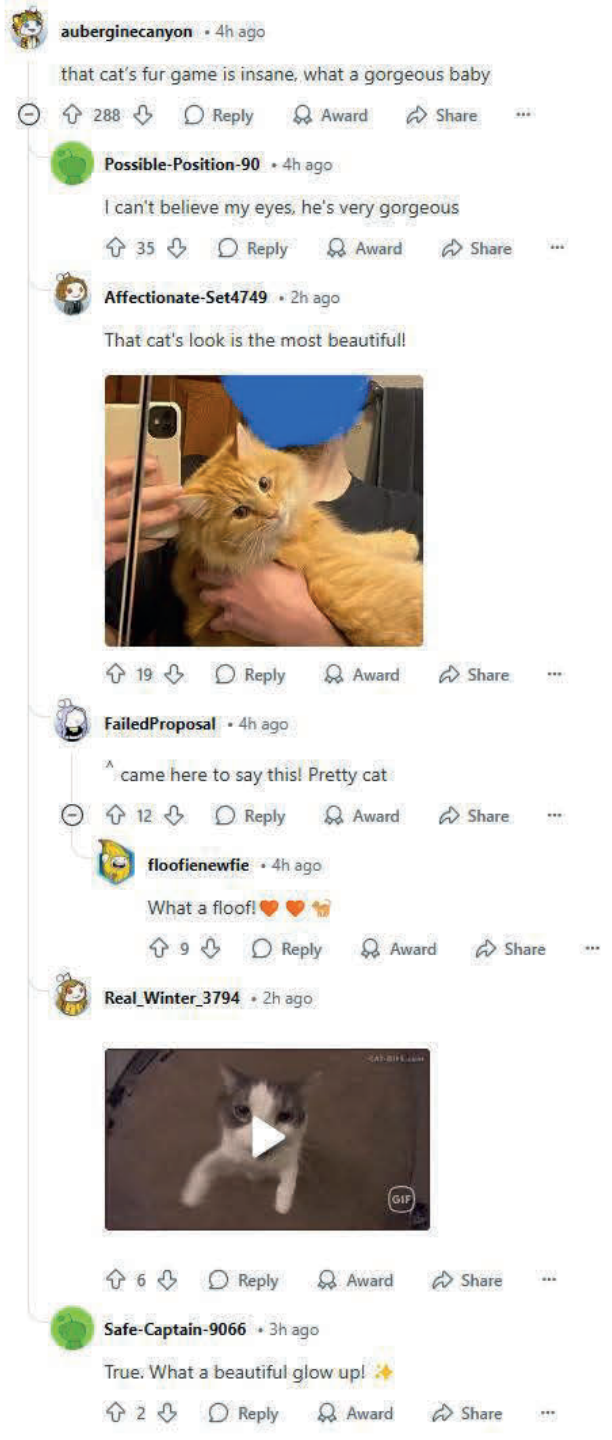
Lazy loading

- Common performance improvement
- An object's children (replies) are lazily loaded/fetched on-demand
- Clicking on '+' loads the remaining replies



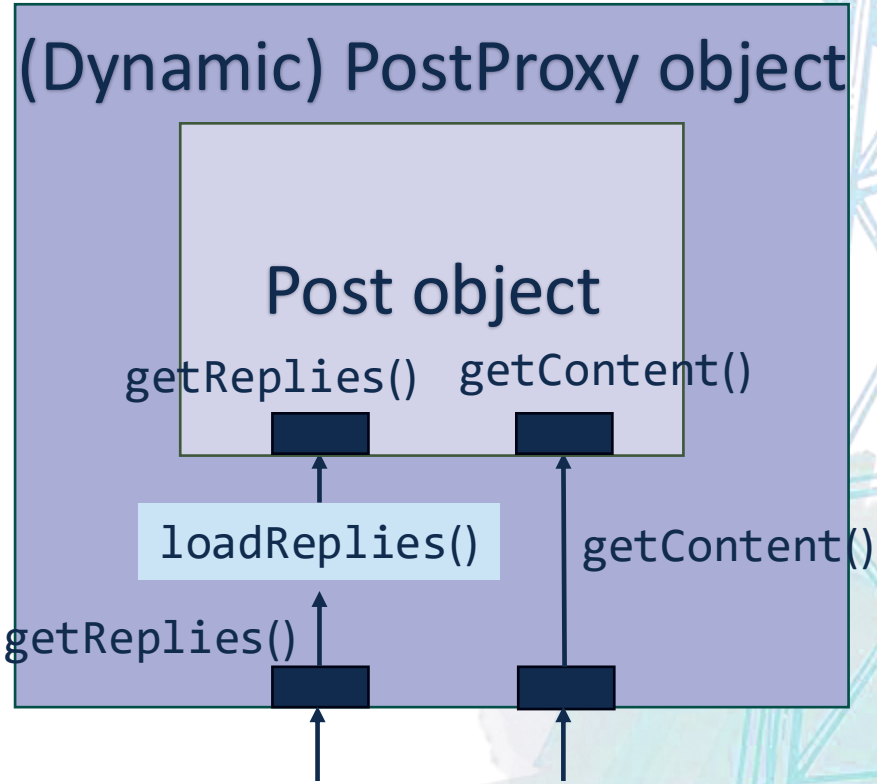
Lazy loading

- Clicking on ‘+’ loads the remaining replies
- Dynamic proxies simplify implementing lazy loading



Proxies for lazy loading

- High level approach
  - Create reply Post objects with only the `postId` (do not load the data using `hgetall()`)
  - Create a (dynamic) proxy for the Post object and have it intercept `getReplies()` method invocation
  - Only when the `getReplies()` method is invoked, perform `loadReplies()` to load all the reply post objects



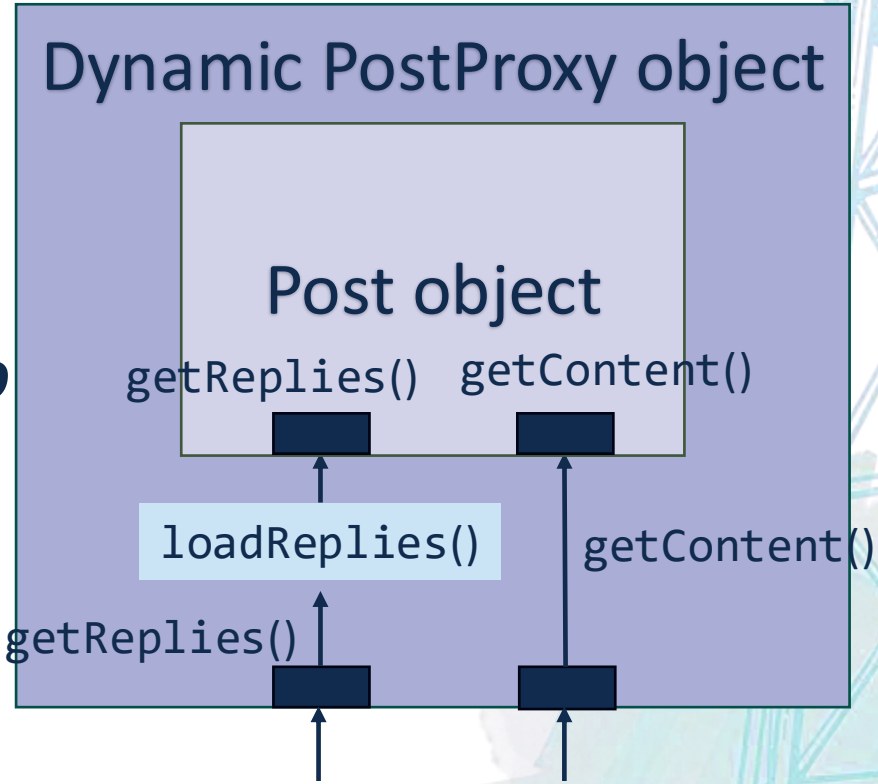
Proxies for lazy loading

```
Post loadPost(String id) {
 map = jedis.hgetAll(id);
 Post post = new Post();
 post.setId(id);
 post.setCreatedAt(map.get(“createdAt”));

 List<String> replies =
    map.get(“childPosts).split(“,”);

 for (String replyId: replies) {
    Post reply = new Post();
    // does not load the reply details
    reply.setId(replyId);
    post.getReplies().add(reply);
 }
 return createLazyLoadProxy(post);
}
```

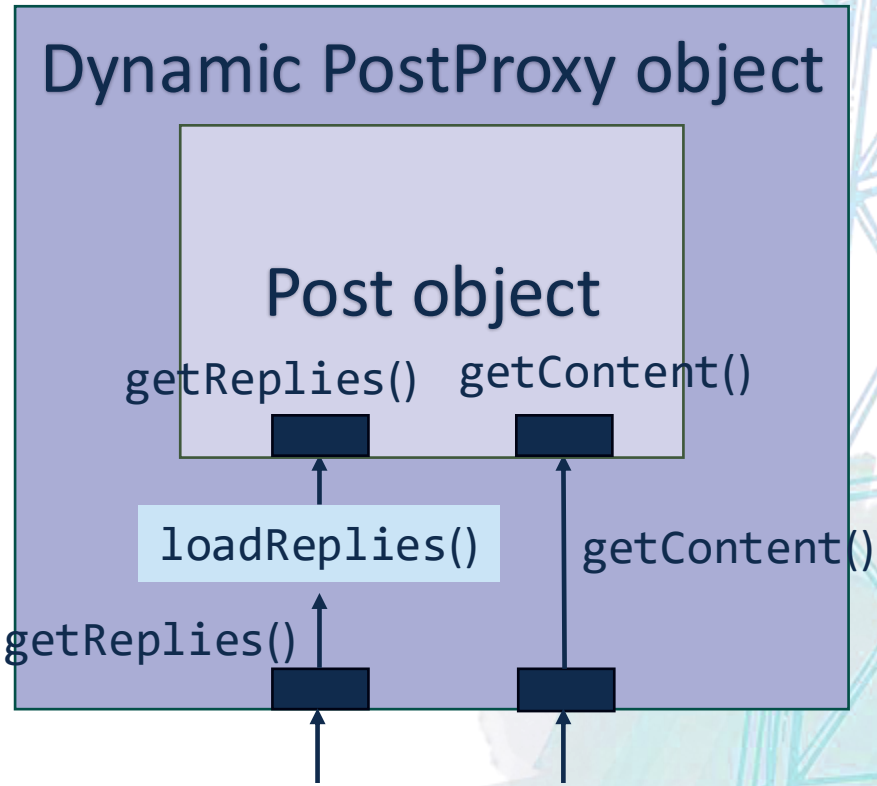
Only set the Reply ID



Proxies for lazy loading

```
List<Post> loadReplies() {
    List<Post> newReplies = ...;
    for (Reply reply: this.getReplies()) {
     // only the id is populated
     Reply reply = loadPost(reply.getId());
     newReplies.add(reply);
    }
    this.setReplies(newReplies);
}
```

Proxy calls loadReplies transparently



Lazy loading code is provided by the proxy class

Reflection + annotations + dynamic proxies

- Can combine proxies with reflection and annotation
- Lazy loading only for annotated fields
  - Lazily load the @LazyLoad annotated fields only when they are accessed
- ***Transparent*** lazy load from the programmer's perspective

```
class Post {
    @Persistable
    Integer postId;
    @Persistable
    String postContent;
    @Persistable
    @LazyLoad
    List<Post> replies;

    // getters and setters for postId,
postContent
    List<Post> getReplies();
}
```

Lazy loading

- Database records
  - All ORMs such as Hibernate support lazy loading using annotations
- File content
  - Lazy load 1 GB file
- Content to be fetched over the network
  - Lazy load remote content



Dependency Injection



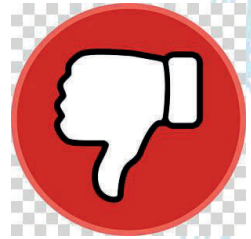
Dependency injection design pattern

- Dependency injection is a design pattern where an object's dependencies are **provided externally** rather than created internally
- Constructor or setter methods
- Promotes loose coupling, modularity, better testability
- `OrderService` depends on an interface and concrete implementation is chosen by the caller

```
// Without DI
class OrderService {
   private PaypalProcessor p = new
PaypalProcessor();
    // -- snip
}

// With DI
class OrderService {
    private PaymentProcessor p;

    OrderService(PaymentProcessor p) {
        this.p = p;
    }
    // -- snip
}
```





Dependency injection design pattern

- Dependency injection is a design pattern where an object's dependencies are **provided externally** rather than created internally
- Constructor or setter methods
- Promotes loose coupling, modularity, better testability
- `OrderService` depends on an interface and concrete implementation is chosen by the caller

```
interface PaymentProcessor { // --snip }
class PaypalProcessor implements
          PaymentProcessor { // --snip}

class OrderService {
    private PaymentProcessor p;

    OrderService(PaymentProcessor p) {
        this.p = p;
    }
    // -- snip
}

public static void main(..) {
    PaymentProcessor p = new PaypalProcessor();
    OrderService service = new OrderService(p);
}
```

Inversion-of-control and DI containers

- DI container is the framework responsible for instantiating and injecting the dependencies
- Inversion of Control means a class gives away control of instantiating its dependencies to the IoC framework
- Goal: design IoC framework that can automatically inject the `A`, `B`, `C` dependencies into `OrderService`

```
class OrderService {
    private A a;
    private B b;
    private C c; // … and more

    OrderService(A a, B b, C c, …) {
        this.a = a; // -- snip
    }
    // -- snip
}

public static void main(..) {
    A a = new A();
    B b = new B();
    // …
    OrderService service = new OrderService(a,
b, c, …);
}
```

Spring IoC framework

- Provides automatic dependency injection
- `@Autowired` componets are automatically injected by the framework
- Provides `@Component, @Service, @Repository` annotations to register ***beans*** which are ***auto-injected***
- Question: how does Spring achieve this?

```
@Service
class OrderService {
    @Autowired
    private A a;
    @Autowired
    private B b; // … and more

    OrderService() {
        // -- no need to assign anything
    }
}

@Component
class A {}

@Component
class B {} // -- snip

public static void main(..) {
    ApplicationContext ctx = // -- snip;

    OrderService service =
       ctx.getBean(OrderService.class);
}
```

Proxies for mock testing



Mocking overview

- What is mocking?
  - Simulate the behavior of real objects in a controlled way
- Useful during unit-testing
- Why?
  - Isolate components for unit testing
  - External systems (APIs, Databases)
  - Complex or time-consuming operations



Mocking overview

- Example: unit test `composeEmail`
  - But don't want to actually send an email to a client!!

```
public EmailService {
    public boolean sendEmail(...) {
        // send an email
        if (success) return true;
        return false;
    }
}
```

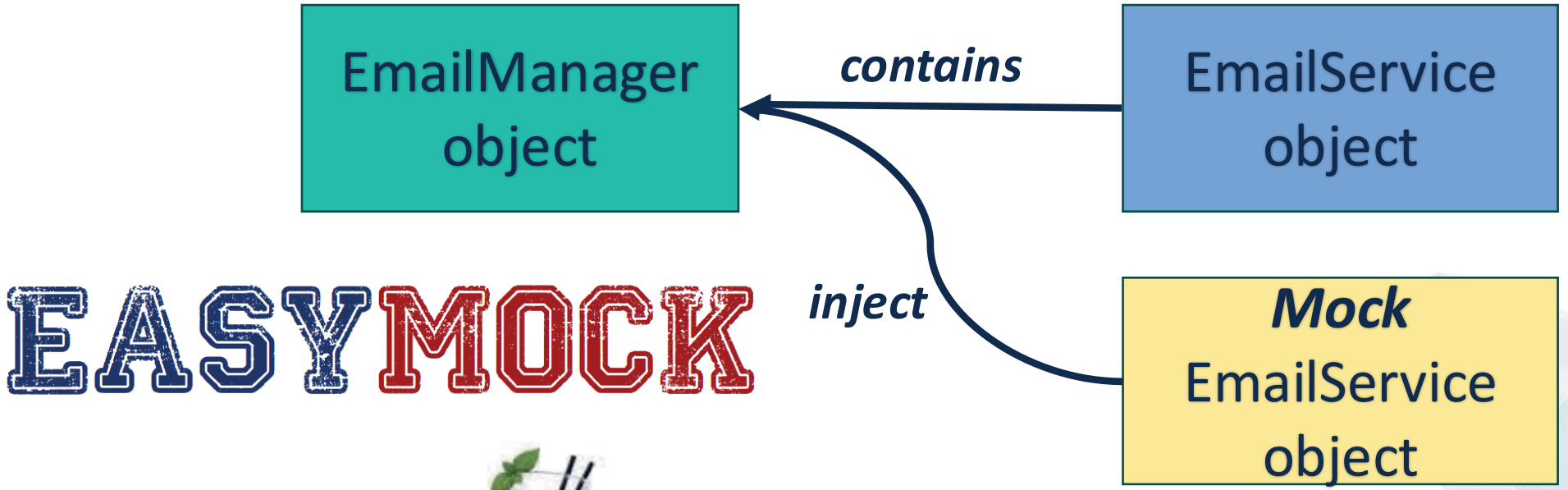
```
public class EmailManager {
    private EmailService emailService;

    private void formatEmail(String email) { … }
    private void displayError(boolean succ) { …}

    public String composeEmail(...) {
        String email = ...;
        formatEmail(email);
        boolean success = emailService.send(email);
        displayError(success);
    }
}
```

Mockito overview

Allows programmer to inject mock objects to ***stub*** functionality



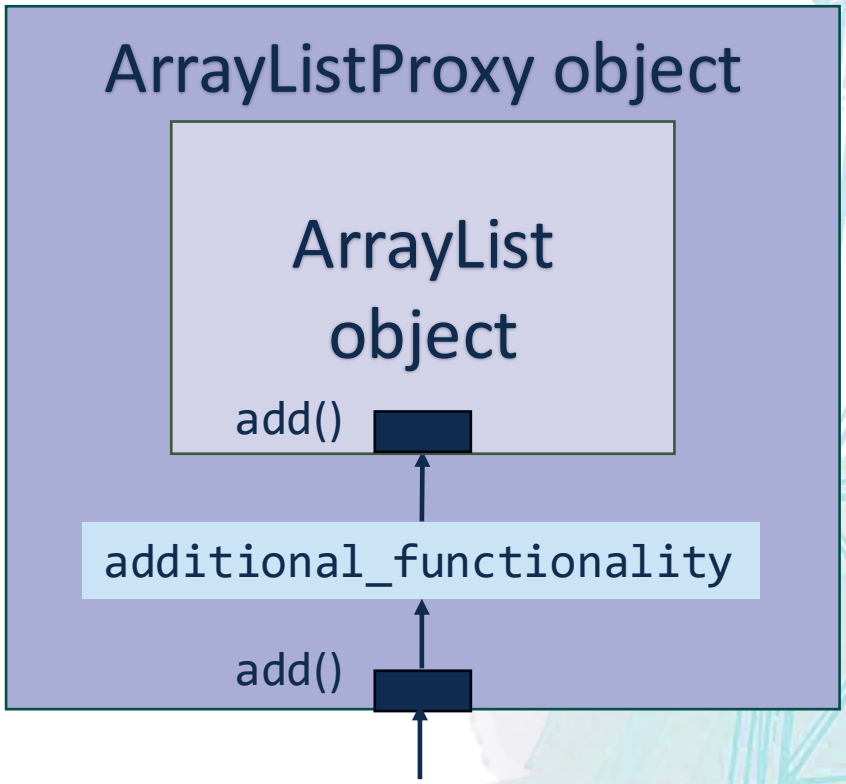
EASYMOCK

mockito

Stubbed `sendEmail()` ***function that does nothing***

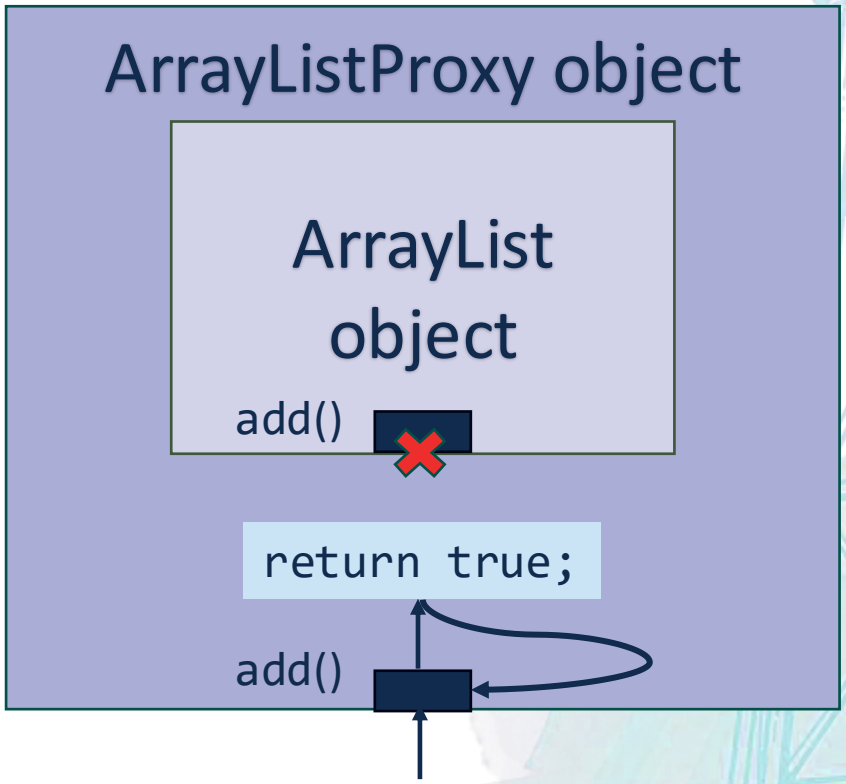
Proxy can hijack functionality

- Previous cases proxies augmented the target object functionality



Proxy can hijack functionality

- Previous cases proxies augmented the target object functionality
- It can also hijack functionality and not invoke the target object's method
- ArrayList proxy object can always return true for `add()` invocation



Proxy can hijack functionality

- Previous cases proxies augmented the target object functionality
- It can also hijack functionality and not invoke the target object's method
- ArrayList proxy object can always return true for add() invocation

```
Object createProxy(Object object) {
 // -- snip

 MethodHandler methodHandler = new MockHandler();

 // -- snip
 return proxyObject;
}

class MockHandler extends MethodHandler {
@Override
    public Object invoke(Object self,
        Method thisMethod,
        Method proceed,
        Object[] args) throws Throwable {
        log("accessing method" +
thisMethod.getName());
        return proceed.invoke(self, args);
        return new Boolean(true);
    }
}
```

Mock an ArrayList

- Mockito's `mock()` method injects a proxy
- Configure the proxy object
  - `when`
  - `thenReturn`

```
import java.util.List;

import static
org.mockito.ArgumentMatchers.anyInt;
import static org.mockito.Mockito.mock;
import static org.mockito.Mockito.when;

public class MyApp {
    public static void main(String[] args) {
        List<Number> myList =
mock(ArrayList.class);
        when(myList.add(10)).thenReturn(true);

when(myList.add(anyInt())).thenReturn(false);

        System.out.println(myList.add(30)); //
return false
        System.out.println(myList.add(10)); //
return true

    }
}
```

Mock an ArrayList

- Mockito's `mock()` method injects a proxy
- Configure the proxy object
  - `when`
  - `thenReturn`
- Mockito proxy objects record method invocations the first time
  - Then replay the configured return value

```
import java.util.List;

import static
org.mockito.ArgumentMatchers.anyInt;
import static org.mockito.Mockito.mock;
import static org.mockito.Mockito.when;

public class MyApp {
    public static void main(String[] args) {
        List<Number> myList =
mock(ArrayList.class);
        when(myList.add(10)).thenReturn(true);

when(myList.add(anyInt())).thenReturn(false);

        System.out.println(myList.add(30)); //
return false
        System.out.println(myList.add(10)); //
return true

    }
}
```

Writing a JUnit test with Mockito

```
public EmailService {
    public boolean sendEmail(...) {
        // send an email
        if (success) return true;
        return false;
    }
}

public class EmailManager {
    private EmailService emailService;

    private void formatEmail(String email) { … }
    private void displayError(boolean succ) { …}

    public String composeEmail(...) {
        String email = ...;
        formatEmail(email);
        boolean success = emailService.send(email);
        displayError(success);
    }
}
```

```
import ...;

public class EmailManagerTest {
    @Mock
    private EmailService emailService;

    @InjectMocks
    private EmailManager emailManager;

    @BeforeEach
    public void setUp() {
        MockitoAnnotations.openMocks(this); // Initialize
mocks
    }

    @Test
    public void testComposeEmail() {
        // Stub the sendEmail method to return true

when(emailService.sendEmail(anyString())).thenReturn(true);

        // Call the method under test
        String result = emailManager.composeEmail();
        // do any assertion checks

    }
}
```

Summary

- Reflection, annotations, and dynamic proxies are very powerful
  - Must be used judiciously
  - Typically, not used in regular application development
  - Used in framework development



Summary

- Frameworks using reflection, annotations, and dynamic proxies are widely used
  - Beneficial to know how they work under-the-hood
  - Very helpful in debugging

```
@Entity
class InsuranceClient {
    @Id
    private Integer insuranceClientId;

    @Column(name = “address”)
    private String address;

    @OneToMany(fetch = FetchType.LAZY)
    List<Policy> policies;

    // getters and setters

}

InsuranceClient client = …;
client.getPolicies(); // sometimes takes seconds
                      // Why?
```

Summary

- Reflection and proxies have application way beyond just Redis persistence and logging
- Used in microservice frameworks, MVC frameworks, dependency injection, database persistence, and so on, in many languages







